

July 13, 2025

Dear Dr. Roberts,

Thank you for the opportunity to submit a revised version of our manuscript for consideration in *PLOS Biology*. We appreciate the time spent by the reviewers and editors on our paper and believe their comments have helped us greatly improve it. In particular, we have streamlined the text, reperformed model optimization, removed analyses that were not core to our main aims, and we have tried to make those aims (along with details of the model and analyses) more clear. Below you will find the reviewer's comments (**in bold**) and our responses to them (line numbers refer to the version with tracked changes).

We hope you agree that this version thoroughly addresses the reviewers' reports, and we welcome any further feedback.

Sincerely,
Avril Wang, Megan Greischar, Nicole Mideo

Reviewer #1:

There is an intrinsic trade-off between investment in with-host growth versus transmission in malaria parasites. This theoretical study explores the influence of different scenarios on the evolution of plasticity in malaria parasites. Following previous studies, the fitness of the malaria parasite is expressed as the cumulative number of transmissible stages of the parasite (gametocytes) over the duration Δ of the infection. Previous theory assumed that the parasite could modulate the investment in transmission as a function of the time since the initiation of the infection. But it is unclear if the malaria parasite is able to track time since infection. The present work explores the evolution of parasite plasticity when the parasite is unable to track time but can track other cues like the density of different types of cells. The authors show that when the parasite has access to a single cue, some cues are better than others. Plasticity based on a single cue, however, yields much lower fitness than plasticity based on time. Next, the authors consider that the parasite has access to two different cues. Using two cues increases the fitness of the parasite. Yet, when the authors introduce some variability among hosts, the optimal strategy based on a single cue may outperform the optimal strategy based on a two cues. They interpret this effect as a lower robustness of the plasticity based on two cues (versus a single cue). Finally, the authors discuss the emerging link between plasticity and parasite virulence in their model and in *Plasmodium chabaudi*.

Major comments:

There are many interesting ideas in this study. But the analysis relies exclusively on numerical approaches and some steps in the derivation of the results are unclear to me.

First, I do not understand the way plasticity is modelled. From equation (7) the investment in transmission depends on some quantity α days ago and uses a linear combination of Heaviside step functions. But a plot of this function would help a lot to illustrate the general shape of plasticity as a function of the cue. Is this a continuous function? Can it take any shape? Besides, I do not

understand how the function can react differently to the density of the log(density). If there is a threshold value above which there is a switch to a different strategy (this is what step functions are modelling) using the density or the log(density) should be the same. But I am probably misinterpreting the way plasticity is modelled (a figure explaining what is done would help). Figure 2 seemed to provide this information but I don't understand panel B and what are the parameters that are evolving to modify the shape of the reaction norms.

We would like to thank the reviewer for their thoughtful comments that have strengthened the paper. The suggestion to include a figure explaining how we modeled and optimized the reaction norms is a great one, and we have implemented it in a new panel in Figure 1 (Fig 1C). Through this figure and in the text, we have clarified how plasticity is modeled, e.g., in the results, l. 110-114,

“For each putative cue, we define its associated reaction norm using flexible splines (Fig 1C). Specifically, the shape of the reaction norm is determined by a linear combination of three basis functions comprising a cubic spline. Adjusting the weight of each function (i.e., the model coefficients) changes the shape of the reaction norm.”

See also l. 599-611 and the caption for Figure 1 for additional information on how we have parameterized and optimized the reaction norms.

Our method of using cubic splines ensures that the reaction norm is continuous. Representing a sharper curve with this method, which is necessary for a reaction norm that changes drastically at a certain threshold, either requires transforming the input variable (e.g., log-transformation) or placing internal knots (which are like the internal boundaries of a piecewise function). We have chosen the former option, given the difficulty of “knowing” the appropriate location to place internal knots. We have redrawn Figure 2B, which shows optimal reaction norms for different cues, with the x-axis on a log scale. This redrawing better emphasizes the differences that exist for reaction norms based on raw versus logged cues. (Perhaps as intuited by the reviewer, the reaction norms look essentially identical for RBC density as a cue, since this variable generally spans less than one order of magnitude in simulated and real infections.)

We note that the use of the Heaviside function only serves to constrain the ranges of the “perceived” cue values within the plausible range and does not play a role in shaping the reaction norm within the defined cue range. We have reworded the explanation of the Heaviside function in l. 613-618.

What are the black and pink bars supposed to mean on the plots?

The black and pink bars are rug plots, which represent the cue values that are “traversed” in a simulation when the parasite is adopting the optimal transmission strategy. This rug plot indicates the approximate range of cue values relevant to a typical infection. In redrawing Figure 2B, we have made it look less busy, and the rugs are more visible. We have also edited the caption to specifically point to the “horizontal bars” that are the rug plots.

Is there an intrinsic constraint linking $c(t)$ and $c(t+a)$ (for any duration of time a)? If there was, I think it would mean that even though your function assumes that "parasites sense only the magnitude of a cue" (line 473), this function may perceive the shape of the variation of the cue.

Maybe this is why the function that uses the log(density) is performing better? The authors need to be clearer about this key step of their model.

Since c is a function of some other within-host variable (for all cues other than time), there is only an indirect link between $c(t)$ and $c(t+a)$. For example, if the cue is RBC density, then the function c must be continuously differentiable with respect to RBC density – a constraint imposed by the cubic spline. However, transmission investment could vary greatly from one time step to the next if RBC density changed drastically across time steps. The rate of change in cue value is, of course, also constrained (or determined) by our within-host dynamical model, imposing a maximum rate of change. We are not sure that this indirect constraint helps to explain our model or results, and we are confident that the $\log(\text{density})$ performs better for parasite-centric cues because it allows parasites to delay transmission investment right at the start of infections. This difference in performance with logged versus raw cues at low parasite densities is more clear with our updated Figure 2B.

Second, some results are very surprising. Why do you get higher fitness with the best single cue model than with the best dual cue models in figure 3A. If one cue was useless, the parasite should evolve not to use it, and thus the two cue models should always yield fitness $>$ or $+$ to the single cue model. I wonder if the fact you use different time step size (line 664) could explain that. Because of the above two points I am unsure about the results presented in the paper. The difference between the use of the $\log(\text{density})$ is key but I don't understand how the authors got there. I like the idea that heterogeneity among hosts may induce a higher fitness cost if the parasite uses two cues but I don't know if this result is real or if there is a "problem" in the way fitness is computed.

We thank the reviewer for noticing this surprising result, and we agree that the dual cue model should always outperform the single cue model. Despite our global+local optimization approach, this result suggested that our optimization routine for dual cues sometimes failed. To remedy this, we re-ran our dual cue optimizations using another approach (methodology detailed in l. 708-730). Briefly, we first attempted to find a dual cue strategy that resembles the best single cue by essentially ignoring the second cue. We then set this strategy (i.e., the parameters describing the spline surface) as the starting point and used the L-BFGS-B algorithm to perform local optimization. Using this approach to optimization, we were able to increase the fitness of the dual cue models, resulting in all dual cues now having a higher fitness than the single cue models, as expected.

The within-host model is trying to capture many details of the biology of malaria parasites. Do you need all these details to develop your argument? My impression is most of the results you are highlighting should hold in simpler versions of your model, and with a simpler model you may be able to get a deeper understanding of the pros and cons of different forms of plasticity. More generally, because this work is very numerical, a journal like PLOS comp biol would be more appropriate than PLOS biol.

We appreciate the reviewer's point that numerical approaches are often undervalued, despite the need for such methods when models consider even ubiquitous aspects of ecology like resource limitation or top-down controls (such as via immune clearance). For example, Koella & Antia (1995, *Theoret Pop Biol*) examined plastic transmission investment using a general model of infection and immunity, an

approach that required numerical simulations despite strong simplifying assumptions like resource-independent parasite proliferation and perfect information about the time until host death or immune clearance. Thus, our model would need to be considerably simplified to become tractable to analytical methods like optimal control. Though predictions about the pros and cons of different forms of plasticity could be made from such a simplified model, it is not clear how those predictions could be generalized to diverse organisms (or to malaria parasites), all of which are constrained by limited resources. Further, previous *PLoS Biology* articles suggest the journal places value on the use of strictly numerical methods to understand system-specific evolutionary pressures, including in malaria parasites (e.g., Bushman et al. 2018). Here we go beyond numerical methods by ground-truthing inferences drawn—specifically about the features of the best dual cue combinations—with empirical data.

Nonetheless, we take the reviewer’s point that streamlining complexity where possible will enhance the impact of this study, and we have attempted to do so, e.g., by reducing the number of dual cue combinations examined and greatly shortening the semi-stochastic modelling section to focus on key takeaways.

Minor comments:

I don't understand the logic of equation (19). Why the difference in the denominator?

In streamlining our manuscript, this analysis has been removed. Though for completeness, the equation being referred to estimated the proportion of the deviation in fitness observed in our semi-stochastic model that could be attributed to a single parameter. The denominator represented the fitness deviation observed when all six parameters were varied simultaneously compared to the deterministic model, while the numerator was the fitness deviation observed with only one parameter varying. We removed this analysis since it was not core to the takeaway that there is a tradeoff between accuracy in a given environment and susceptibility to noise.

Some discussion about the ability of the malaria parasite to perceive the density (or the log(density) of cells (infected or not) is needed. The model relies on this ability and it is important to discuss the feasibility of this assumption. What type of signal could carry this information?

We thank the reviewer for this comment, which made clear to us that we had failed to establish in the introduction what is known empirically about the ability of *Plasmodium* parasites to detect and respond to their environment. We have substantially revised the introduction, and this information is now included, l. 28-38,

“Importantly, transmission investment in Plasmodium spp. is plastic, changing in response to conspecific density [11, 12], replication rate [13], parasite-conditioned media [14, 15] (which could contain infected red blood cell (iRBC)-derived vesicles [16]), RBC age structure/availability [13, 17–19], and antimalarial exposure (e.g., [13, 20]). Understanding why transmission investment is plastic and what the optimal plastic strategy is has been the subject of several prior studies [7, 13, 21–24]. In one example, using the experimentally tractable rodent malaria system, Plasmodium chabaudi, and different doses of antimalarial drugs, Schneider et al. showed that intermediate reductions in parasite density led to reduced investment in transmission

while strong reductions had the opposite effect. These results are in line with standard predictions from life history theory...”.

We have also added related text to the Discussion, e.g., l. 376-381,

“Prior studies have found that Plasmodium transmission investment varies in response to conspecific density [11, 12], parasite-conditioned media [14, 15], and infected red blood cell (iRBC)-derived vesicles [16], suggesting that parasites possess the ability to sense iRBC density. Our results support these responses being part of an adaptive strategy.”

Reviewer #2:

Wang et al. studied the intricate timing of investment into reproduction in malaria parasites. Specifically, they examined the types and number of cues sensed by the parasites' sexual reproduction investment to maximize their fitness. They found that dual cues instead of single cue are required to realize the terminal investment and logged cues are necessary for the delayed investment. Dual cue sensing also increases disease virulence. However, dual cue perceptions have tradeoffs with the fitness cost due to environmental stochasticity. Overall, I think the manuscript is very well written. I have two general questions regarding the model assumptions and some minor comments.

We greatly appreciate the reviewer's feedback, which has significantly improved the paper's readability.

Q1: It's not immediately clear to me how the time-varying optimal investment strategy is derived (Fig. 1A, Fig. 7). In the "fitness optimization" section, it only documented how the optimal trajectory is derived using single or dual-cues, but not for the scenario when investment is a function of time directly (as investment due to responses to the cues are delayed in time).

We thank the reviewer for bringing this omission to our attention. We have now made explicit in the methods how we do this (it is essentially the same as the single cue models, except no Heaviside transformation is required), l. 620-622,

“For the best time-varying reaction norm, we defined the perceived cue as the time since infection, which has a range of 0-20 days and a step size of 0.001 (same as the simulation time step). In this case,

$$c(t) = \exp(-\exp(B3(t))).”$$

Q2: When investigating the impact of environmental stochasticity on the optimal transmission investment, the authors assume the parasites use the same optimal strategy inferred from the deterministic model under varying changes of parameters, such as burst size and immune activation strengths. Is that a reasonable assumption? If the optimal strategy is a result of long-term selection from host-parasite interactions, the outcome would be the strategy that performs overall the best under varying parameter combinations instead of a specific strategy trained from the deterministic model. I wonder if a different optimal strategy can be trained while allowing for realistic variation of the six parameters.

This comment is related to another from Reviewer #4, who identified our implicit assumption that parasites switch hosts very quickly. We now make this assumption explicit, l. 253-257,

“Specifically, the optimal reaction norms identified above are associated with a particular set of model parameters (the median estimates from posterior distributions in [41]; we refer to this as the deterministic model). This implicitly assumes that hosts are all the same or that parasites are moving between hosts sufficiently quickly that they adapt to an “average” environment.”

and, l. 271-273,

“Here we are asking, if parasites are “optimized” to the average host, does environmental and developmental variation change inferences about which cues are best”?

Although it would be answering a different question than the one we have posed, it is in principle possible to find the optimal strategy in the face of variable parameters. Technically, this would be challenging for a few reasons. First, if parameters are chosen randomly over a series of infections, this will make convergence of the optimization algorithm difficult. Second, if instead a specific series of parameter sets is chosen, then more assumptions need to be made (e.g., How many parameter sets? Are infections with each parameter set equally likely?). We note that with enough parameter sets, chosen with weights defined by the posterior distributions that we have relied on, we’d expect to find the same optimal strategies as using the medians of those distributions, as we have done. Thus, we think the assumption we have made is a reasonable place to start.

Also, I'm a bit confused by why "burst size" or host immune responses are considered environmental factors. Parasite burst size is determined by parasites' genetic factors and host-parasite interactions, which are part of the overall parasite sexual and asexual reproductive strategy that is also under selection.

We have defined the host immune response as an environmental factor, given that the immune system is an aspect of the environment experienced by the parasite. We agree that parasite genetics will have a large influence on burst size, though environmental influences have also been shown. Specifically, Stürmer *et al.* (2023, *PLoS Pathog*) recently demonstrated *in vitro* that nutrient level mediates burst size in *P. falciparum*. We have amended our language throughout to capture more sources of variation (e.g., “environmental and developmental stochasticity”).

Minor comments:

- Fig 2B, legend label, "none" should be "raw". More explanation for Fig 2 is required: It seems that when using logged cues, transmission investment is very high for low densities compared to raw cues (Fig. 2B), this seems to be counterintuitive with the fig 2A result when delayed responses are observed with logged cues.

We have corrected the Fig 2B legend label to “raw” and edited the caption for clarity. In light of this comment, we have also redrawn the plots in Fig 2B, with log-transformed x-axes for easier visualization at low densities. The revised plots better illustrate that transmission investment is indeed lower for the logged cue at low densities of parasite-centric cues.

- Lines 237-243, seems to be discussion contents instead of results.

We agree. This text has been shifted to the discussion section (l. 466-474).

- Fig 5. The legend says fitness from semi-stochastic model is rarely larger than that from the deterministic model, but it's not immediately clear from violin plots of Fig 5B, especially for burst size.

In revising our paper to make it more streamlined and clear, we decided that this figure and its associated analysis (determining which parameter contributed most to the loss of fitness in the semi-stochastic model) was not central to our main message. Thus, we have removed this figure and replaced it with another that is responsive to other reviewer comments, illustrating the tradeoff between accuracy in a given environment and susceptibility to (or precision in the face of) noise.

- Fig 6. The meaning of colors and lines in B-C are not explained

We have similarly removed Figure 6 in streamlining the article. We note that the bottom panel of our new Figure 5B is similar (though not identical) to the previous Figure 6B&C, showing the consequences for transmission investment when particular cues are perceived in the face of environmental or developmental stochasticity. (We now also show the scale of the variation in those cues from the same simulations in the top panel of Figure 5.) We have been careful to describe the different coloured lines in the Figure 5 caption.

- Fig 7. Why are the regression lines required here? Seems to be unrelated to the main message you are conveying. What do the grey points represent in the figure?

Good point — the lines weren't required and we have removed them from the figure (now Figure 6 in the revised version). The grey points represent all cues that we have not colour-coded in the legend. We have added additional clarification in the figure legend to explain this.

Reviewer #3:

Wang et al present an interesting theoretical model focused on understanding the role of cues in reproductive investment using the well studied rodent plasmodium model system as a case study. The study presents a significant advancement in understanding how malaria parasites regulate transmission investment in response to their within-host environment. The authors successfully integrate life history theory with mathematical modeling, providing a robust theoretical framework that aligns with empirical observations of transmission investment plasticity. Their results are a major step forward because they relax the assumption that parasites operate with perfect knowledge of infection time and instead propose a more biologically plausible mechanism based on environmental cue perception.

One of the paper's key strengths is its rigorous theoretical approach, which systematically evaluates various putative cues and their impact on parasite fitness in a well understood system. The distinction between single and dual cue perception is particularly insightful, demonstrating the advantages of state differentiation in guiding reproductive decisions. The finding that dual cue perception increases virulence also aligns with broader evolutionary theories that link optimal resource exploitation with pathogen fitness.

The key focus on the impact of different cues on the ability of the parasite to get more optimal outcome is very original. This is a strong reason to publish the paper, since it is a major and general conceptual advance. We don't normally think about how information on environment is some sense optimally used for organisms to optimize their fitness. It's a very general problem that has not been sufficiently addressed. Also the paper is focused on reproductive effort which is a fundamental trait and moreover using the rodent malaria system is excellent since the investment in reproduction as opposed to growth is very clearly defined. Much more so than in most systems. This allows a really comprehensive analysis. Finally the fact that they are studying malaria means that they can relate their results to important practical questions such as the evolution of virulence.

Overall, I think this is a tour de force of a paper, original in concept and extremely rigorously carried out. My criticisms emerge from just how detailed it is which results in a slightly overwhelming presentation of results and details. I think the paper would be strengthened with more discussion of the key result that multiple cues are needed for the parasites to recognize the need for terminal investment. This is really interesting and although it is discussed, I think an expanded discussion would help people understanding this intuitively. There are a lot of interacting parts here in terms of cues, and the complexity of the models, so perhaps some more intuitive discussion of two of the cues individually and why they don't work on their own but do when combined would illustrate the points better. The paper is very long and perhaps pulling this key discussion into a conclusions section would help with the structure.

We would like to thank the reviewer for their encouraging feedback. In response to this comment and others, we have streamlined our manuscript, removing details and analyses that were not crucial to our main messages, and have offered more intuitive explanations whenever possible.

Another interesting advance in the paper is the discussion of emerging tradeoffs involved in multiple cue perception. Again I think the paper would be improved with a more intuitive overview of how these trade-offs emerge and where they are underpinned by assumptions. It makes sense that while the dual cue perception improves parasite fitness and enables more precise transmission strategies, it also increases susceptibility to environmental and developmental fluctuations, but I think it would have a greater impact if people can see where this emerges from intuitively.

We agree with the reviewer that the paper needed editing to provide more intuitive understanding throughout, and we hope we have achieved that. With respect to the fitness advantage of dual cue strategies and the resulting tradeoff that emerges, we have done the following. First, we have added a new paragraph at the beginning of the dual cue results section (l. 156-166), and a new panel in Figure 3, to give readers an intuitive understanding of why single cues fail to reproduce the expected optimal pattern

of transmission investment, based on the dynamics of those single cues. This sets up the expectation that two cues will be able to recreate more aspects of the optimal investment pattern and motivates the question of which set of cues is best.

Second, we have streamlined the section on environmental and developmental stochasticity, dispensing analyses that were not core to understanding the tradeoff (e.g., exploring which model parameter drives fitness losses in variable environments). With the focus of this section now squarely on dual cue strategies being more susceptible to noise, we again appeal to the dynamics of the cues to show the reader how and why this tradeoff emerges. We do this through a new paragraph at the end of that section (l. 290-303) and an updated Figure 5B, which demonstrates that variation in cue trajectories over the course of an infection gives rise to wildly different patterns of transmission investment in the dual cue, but not the single cue case. This is because deviations from the median dual cue trajectory traverse qualitatively different paths through the finely-tuned dual cue reaction norm surface (shown in the top panel of Fig 5B).

Another important result that could be highlighted more is that, dual cue perception correlates with higher virulence, as it allows parasites to exploit hosts more effectively, potentially leading to more severe infections. This has implications for understanding parasite evolution, within-host competition, and malaria transmission dynamics, particularly in response to interventions like drug treatment or vaccines. Again I think the underlying assumptions and intuitive response for this result could be highlighted more in order to make this result pop.

We agree that this is an important result! We hope that by streamlining and simplifying the section that comes directly before the one on virulence (i.e., the section on stochasticity), that these results will “pop” more. We have also been careful to define virulence throughout the manuscript and have edited the paragraph in the discussion that relates to these results (l. 420-430).

All of this advice is given in the hope that the paper can have a broader impact beyond the rodent malaria model system community. I think the general insights that can be gained here have the potential to influence more general reproductive allocation and optimality studies. At the moment the paper is very detailed, which is a key strength, this is rigorous well grounded theory, but the intuition behind some of the big results is a little buried. Highlighting this will allow the paper to have a bigger impact. I would suggest a major rewrite of the discussion with either new paragraphs that give specific examples that allow the intuitive explanations for why individual cues don't lead to terminal investment and a broader implications outside of the model rodent malaria system, or perhaps an intuitive explanation and wider implication sentence in each paragraph.

We thank the reviewer for this incredibly helpful advice. We hope that by expanding the results section to incorporate more intuitive explanations of key inferences, in combination with removing unnecessary detail and analyses, we have produced a manuscript that is easier to read, with the potential for greater impact. With respect to the discussion, we have made a number of changes. Here we highlight three. First, we provided additional information linking our modelling result on the benefit of perceiving parasite-centric cues to the broader, empirical literature (l. 377-381),

“studies have found that Plasmodium transmission investment varies in response to conspecific density [11, 12], parasite-conditioned media [14, 15], and infected red blood cell (iRBC)-derived

vesicles [16], suggesting that parasites possess the ability to sense iRBC density. Our results support these responses being part of an adaptive strategy.”

Second, we rewrote the discussion section detailing the benefits and drawbacks of sensing multiple cues. To give readers a more intuitive understanding of why dual cue perception is beneficial, we provided a simplified explanation on why perceiving a single cue cannot allow for terminal investment (l. 387-390)

“Since most within-host variables go up and down, or vice versa, over time (e.g., iRBC densities rise and then fall due to resource limitation or immunity), one cue cannot provide enough detail to unambiguously establish “where” a parasite is in the course of infection. Combining cues solves this problem.”

Third, we expanded our discussion of sensing multiple cues in order to make connections to the broader literature on phenotypic plasticity (l. 399-410)

“There is a growing appreciation that, across diverse organisms, plasticity is often expressed in response to multiple cues that allow for more finely-tuned strategies [70–72]. Our work adds support to theories suggesting that the adaptive benefits of phenotypic plasticity are enhanced when cues provide non-redundant information [73] and enable organisms to sense distinct states that correspond to specific optimal phenotypes [74]. Yet it is also widely appreciated that integrating multiple cues presents challenges, including the additional metabolic burden of sensing another cue [75] and the multiplicative effect of noise when cues are non-additive [72]. Our complex (decidedly non-additive) dual cue reaction norms present a trade-off: they allow parasites to more accurately recreate optimal strategies in a given environment while rendering them more susceptible to stochastic environmental and developmental change, i.e., host-to-host variability.”

Minor comments

While the model focuses on *P. chabaudi*, extrapolating these findings to *P. falciparum*, the primary human malaria parasite, requires caution. *P. falciparum* has a longer gametocyte maturation period, which could influence cue reliability and transmission investment dynamics.

We agree with the reviewer that extrapolating our findings to *P. falciparum* requires caution. We have noted the discrepancy between *P. chabaudi* and *P. falciparum* sexual developmental timeline in the discussion and explained how those differences could affect cue reliability (l. 436-440),

*“The human malaria parasite *P. falciparum* has a longer sexual development period of 9-12 days [85] and this longer lag between cue perception and phenotype expression will reduce cue reliability [86]. Thus, we anticipate the value of sensing gametocytes to decrease even further in this species.”*

The model tracks the dynamics of red blood cells (RBCs) and various parasite stages within the host over a 20-day infection period. to identify the best strategies for maximizing transmission fitness, the study employs a combination of local and global optimization algorithms: I think it would be useful to explain the broader uses of the dual methods of the L-BFGS-B algorithm (Limited-memory Broyden-Fletcher-Goldfarb-Shanno with Box constraints) and Global Optimization through the Differential Evolution (DE) algorithm, implemented in the DEoptim

package. This hybrid approach helps identify globally optimal reaction norms and could be used more widely.

We agree that taking a hybrid approach has greatly facilitated our search for global optimums. However, we would like to clarify that the suitable choice of optimization algorithm varies across models and requires empirical determination. Therefore, we are hesitant to recommend the L-BFGS-B and DE algorithms for other optimization problems broadly.

Reviewer #4:

This was an interesting paper that applied a variety of mathematical approaches to explore the question of parasite plasticity in infection, using the reproductive investment of malaria parasites as a case study in this question. The authors find that cue perception mediates parasite life history and fitness and they perform optimization over several hypotheses for cue perception and identify key infection variables that singly or jointly determine the optimal cue. They go even a step further and explore optimal strategies in the context of local vs. global optimum, considering host and infection parameter heterogeneity. I enjoyed reading this paper, and think that it demonstrates interesting biological insights but I am afraid that there is too much going on and some parts need attention, clarification and streamlining. Specific comments are below:

We thank the reviewer for their positive feedback. We agree that there was too much going on, and we believe this revised version is much improved and clearer.

Overall:

This paper relies a lot on results from past work by the same authors, particularly for the model construction and motivation to explore the variable-dependent strategy of reproductive investment. I would like to see more details on the relevant conclusions from this previous work whenever it is cited in this paper, so as to make the paper more self-contained, and highlight more the rationale for the new approach adopted here. Especially in the introduction and beginning of results, e.g. line 82, it seems the reference to previous work is too quick and insufficiently informative. I would really suggest to highlight more the results from the previous work, the use of time as a proxy for "environmental quality" over infection, but a kind of arbitrary proxy, that in different infections could be inaccurate, so there is a basis for exploring concrete infection variables as more informative and parameter-transcending cues.

The reviewer makes an excellent point about time being an arbitrary proxy that is likely to be inaccurate in different infections. We have added text to the introduction to highlight this point, l. 55-63,

“In addition to assuming that parasites can, in effect, “tell time”, this work also assumes all infections proceed identically, a limitation common to models of age-specific reproductive investment [28]. In reality, time would be a poor proxy for environmental conditions since the pace of infection changes with parasite dose [29, 30], host immunity [30, 31], and myriad other

factors. Understanding the links between transmission investment and within-host environmental cues is necessary to validate modelling results with experimental evidence and for predicting the influence of novel perturbations, like vaccination, on infection dynamics and outcomes.”

In response to this comment, we also updated Figure 5A (comparative fitness of cues in the deterministic and semi-stochastic models) to include parasites sensing time. This plot makes clear that under varying conditions, parasites that sense time perform in the bottom half of the cues we tested, a result we now highlight in the text (l. 282-284).

Finally, we also broadened the introduction to rely less on previous theoretical work and include more detail on what is known empirically about the environmental influences on transmission investment in malaria parasites (l. 28-37).

I am not sure also I understand the big aim of this paper. Is the terminal investment the big issue here? The authors are trying to find 'internal' infection mechanisms for terminal investment in reproduction? E.g. the aim articulated in the lines 135-140?

We apologize for having been unclear about our main aims. Our goal was to understand what cues parasites *should* sense when making decisions about transmission investment. While terminal investment is an aspect of the “ideal” transmission investment strategy outlined in prior work, it is not the only aspect that we are looking for. For instance, delaying transmission investment at the start of infections leads to much higher fitness gains than does terminal investment at the end of them. Throughout the manuscript, we have tried to better signpost our aims, e.g., in the abstract,

“We extend that theory to ask which cue(s) should parasites use? Put another way, which cue(s) maximize parasite fitness, quantified as host infectiousness during acute infection?”

and in the introduction, l. 74-84

*“We use numerical optimization to investigate which within-host environmental cue or cues malaria parasites “should” sense for determining transmission investment (i.e., which cues give rise to the highest parasite fitness) and the optimal shape of the relationship between each cue and transmission investment (i.e., the optimal reaction norms). We assess the consequences of perceiving different cues on infection dynamics and disease severity and quantify the robustness of different transmission investment strategies in the face of environmental and developmental stochasticity. Our study further characterizes plasticity in *P. chabaudi* life history, identifies the cues we expect parasites to be using, and introduces a mathematical framework that explicitly incorporates cue perception that could be broadly applied to other organisms.”*

In my view, the paper would benefit from shortening and streamlining and clearer focus. As it is now, a lot of results are presented, they are all interesting, but somehow it's easy to miss the forest for the trees, the findings are all complex and hard to understand together, the model, cue definition, the optimization, the interpretation of the joint cues, the environmental variation, the optimization over environmental variation. The authors could be more selective and decide to divide the big story in two parts that would be able to breathe more and speak more in separation. This is a suggestion.

We completely agree with the reviewer that our original paper was trying to cover too much ground. We decided to remove analyses and text that were not central to our key takeaways, that (i) sensing two cues enables parasites to state differentiate, make better transmission investment “decisions”, and achieve higher fitness, (ii) there is a trade-off between accuracy in a given environment and sensitivity to environmental noise, and (iii) the evolution of plastic transmission investment will influence virulence.

To streamline the paper, we have:

- 1) Focused on a subset of dual cues instead of all possible pairs of single cues (i.e., only combinations of log-transformed cues given their higher performance in the single cue optimizations, see Fig 3B and l. 173-175).
- 2) Removed a figure panel (formerly, 5B) and analysis of which parameter in the semi-stochastic model explained the fitness decline, since it was not central to our takeaway about the tradeoff (ii above). We have replaced that subfigure with another showing the dynamics of the best single and dual cues and their associated patterns of transmission investment, which help explain how and why that tradeoff emerges.
- 3) Removed several paragraphs and another figure (formerly Figure 6) that investigated the mechanism by which dual cue perception sensitizes parasites against environmental stochasticity. We think the intuitive explanation offered by our new Figure 5B is sufficient.

Structure:

- The paper starts with results, but the model is very crucial to understand the results. I wonder if it would be possible to move the materials and methods before the results. Specifically, for unfamiliar readers, the model is not presented in sufficient detail and especially the key mechanisms that drive the end of infection, e.g. the role of resource limitation and immunity. Immunity is not mentioned until much later. This makes it difficult to understand what are the key players mathematically.

We assume that we are limited in restructuring the article, as per PLOS Biol guidelines, although we are open to editorial feedback on this. Nevertheless, we acknowledge the reviewer’s point that there was insufficient detail about the model. In light of this comment, we have substantially revised the first two paragraphs of the results, which describe the model. We have also added a new plot to Figure 1 that illustrates our approach to modeling reaction norm evolution (Figure 1C).

- Some part that is currently in the discussion, e.g. lines 417-428 would be more useful to be put earlier, perhaps in the introduction. Again the rationale for the phenotypic plasticity modeled in this way is important to be highlighted more.

We thank the reviewer for the suggestion and agree that there was insufficient background in the introduction on what is known empirically about *Plasmodium* spp. transmission investment and environmental influences. As noted above, we now include this detail, l. 28-37,

“Importantly, transmission investment in Plasmodium spp. is plastic, changing in response to conspecific density [11, 12], replication rate [13], parasite-conditioned media [14, 15] (which could contain infected red blood cell (iRBC)-derived vesicles [16]), RBC age structure/availability [13, 17–19], and antimalarial exposure (e.g., [13, 20]). Understanding why

transmission investment is plastic and what the optimal plastic strategy is has been the subject of several prior studies [7, 13, 21–24]. In one example, using the experimentally tractable rodent malaria system, Plasmodium chabaudi, and different doses of antimalarial drugs, Schneider et al. [13] showed that intermediate reductions in parasite density led to reduced investment in transmission while strong reductions had the opposite effect. These results are in line with standard predictions from life history theory for malaria parasites...

- I also think the measures that are focus of the investigation, like total transmission fitness or virulence should be stated before the results are presented. It's hard to track the meaning of these different measures, without them being defined.

Great point. We have added the definition of fitness in the first paragraph of the Results, l. 99-102, *“We quantify the fitness of parasites sensing various cues using cumulative transmission potential, which measures the summed probability of infecting mosquito vectors across a 20-day infection period, assuming a constant biting rate and mosquito abundance [23].”*

We note that our operational definition of virulence is given at the start of the section “Dual cue perception leads to higher virulence” (l. 307-311). We are also careful in the abstract and result to equate virulence with infection severity, though we have also made this explicit (l. 315-319, also see response below).

- The model variables also need to be explained in a more reader-friendly manner. For example, start with the uninfected cells, then infected cells, etc. the model without cue- in a clearly-titled subsection of the methods, and then the subsection with the cue highlighting its formulation, and its exact place in the classical model. But please put the equations before and the cue formulation later. As it is now, the cue comes before the other model elements (equations 10-11..).

Thank you for the suggestions. We have rearranged the Methods section such that the model without a cue is described first, followed by a new subsection describing how transmission investment is captured.

Another suggestion is with regards to the order of presentation of the variables and behaviors that define the stages of infection (lines 160 -190) and the optimization (previous paragraphs). Now the optimization is presented first, then the stages of infection as a supporting interpretation. I think for people who read the paper, it would be more useful and feel somewhat more natural to get familiar with the model, infection behavior, stages, etc. and then build an intuitive expectation which leads up to the optimization search and confirms what one would expect from this sort of preliminary sensitivity analysis based on pure study of model behavior and coupling between variables of a dynamical system.

This is a great point and led us to reflect on how to provide the reader with an understanding of the model's behaviour and an intuition about what to expect in the dual cue models. To this end, we have added an additional panel in Figure 3 (Figure 3A) and a new paragraph at the beginning of the dual cue results section, l. 156-169,

“We found that parasites sensing a single cue cannot reproduce the expected optimal transmission investment strategy, particularly the rapid increase in transmission investment during late infection (i.e., terminal investment). Achieving terminal investment presumably requires sensing a cue that provides a unique signal at the later stages of infection, avoiding prematurely increasing investment and reducing fitness. Inspecting infection dynamics produced by parasites that sense time revealed that only RBC density provides such a distinctive late-stage signal (Fig 3A). Given the poor performance of this cue in early stages of infection, these results suggest that simultaneous perception of RBC and a parasite-centric cue could provide infection stage-specific information that enables both early reproductive restraint and terminal investment. Consistent with this expectation, in a rodent malaria experiment, including variables that reflect parasite density and host resources (specifically, replication rate and changes in RBC density) in statistical models explained substantially more variation in transmission investment than the parasite-centric variable alone [13].”

Other comments

- I am a little confused about the way immunity effects are implemented in the model. I have not seen earlier this formulation with the logarithm ($-\ln(1-N(t))$) why not use a simple mass-action term for the interaction between immunity and infected cells? Can the authors please elaborate on the foundation origin and interpretation of this formulation?

We have added further clarification on this formulation, which is borrowed from our previous work (Kamiya et al. 2020 *PLoS Comp Biol*), l. 497-525,

“As in [41], we define N and W as the proportion of cells cleared per day by indiscriminate and targeted responses, respectively...

Because N and W have been defined as proportions of cells cleared, they can be rewritten to

reflect daily rates of clearance. For example, $N = 1 - e^{-x_N}$, where x_N is equal to the rate of clearance on a given day, and is defined as $-\ln(1 - N)$, with some rearranging of terms. A similar expression provides the daily clearance rate for targeted killing, $-\ln(1 - W)$.”

- In the definition of parasite fitness, the probability that the host is alive is not taken into account. Normally the integral would also take into account host survival, e.g. a decreasing exponential, function of total parasite load. Why is this not taken explicitly into account in equation 18? While on the other hand the authors also track the "virulence" but in a decoupled manner from transmission fitness.

The reviewer is correct that we track virulence decoupled from parasite fitness, due to growing recognition that disease-induced mortality may be too rare to serve as an important constraint on the transmission success of pathogenic organisms (e.g., Bull & Luring 2014 *PLoS Path*; Kennedy 2023 *Proc B*). In the context of malaria parasites, the overwhelming majority of infections are subclinical and nonlethal, at least in humans where data on natural infections are available (Lindblade et al. 2013 *Expert Rev Anti-Inf Therapy*). Recent modeling of *P. chabaudi* infections supports the idea that other factors, especially resource limitation, form a more important constraint on parasite fitness than host mortality (Pak et al. 2024 *Evolution*). We now clarify these points in the text, l. 315-319,

*“We note that we are using “virulence” as synonymous with infection severity and not host mortality, thus we assume no link between virulence and the duration of infection. Recent modeling suggests that resource limitation imposes a more important constraint on the evolution of transmission investment in *P. chabaudi* than does host mortality due to infection [61].”*

- I would suggest writing equation 17 in a more general form and simplifying the exponential^(log..) it becomes just $A * G(t)/(1+B * G(t))$ where A and B are just constants and then replace the particular numerical values. It's hard to see its general shape in this current formulation.

We have replaced equation 17 with a more general form, as suggested (now equation 18).

- I am not sure the comparison of the optimized reaction norms with random splines is relevant or particularly informative.

The comparison with random splines serves as a sanity check to ensure our optimization is successful. In the past, when our optimization was not successful, we found that random splines performed better than our “optimized” splines. We have left the associated text (l. 732-737) and supplementary figure (S1) in for now, but are happy to take editorial direction on removing it.

- In lines 580-586, it would be useful to state clearly the optimization framework max(...) over the model coefficients (parameters to be searched).

We now specify in the Methods that the maximum bounds are “defined in Table 3” (l. 616-617).

- In the global optimum part, I am not sure the host heterogeneity that the authors mean, is correctly referred to, when they use the term stochasticity. I would warrant them to consider more carefully the two concepts as environmental stochasticity is one thing, host heterogeneity is another. And it would be very important to define the rate of 'host switching' or 'environmental switching' and see how to local optimum compares to the global optimum. I am not sure the authors did that. In the limit the environment is not switched at all - we have the local optimum, in the other extreme where the hosts are switched very rapidly, there is no point in optimizing to any given host but to the average host. In between, things can be complicated. I wonder if in their simulations with random parameters, the authors are implicitly assuming infinitely fast rate of switching environments.

We thank the reviewer for this insightful comment, which gave us a deeper appreciation for the implicit assumption of our approach. For our deterministic model, most of our parameters are derived from the medians of posterior distributions previously generated (Kamiya et al. 2020 PLoS Comp Biol) and also used in the semi-stochastic modelling. This means that our deterministic optimization can be thought of as optimizing towards the average host when host-switching is fast, or that all hosts are the same. We know make this explicit in the text, l.253-257,

“Specifically, the optimal reaction norms identified above are associated with a particular set of model parameters (the median estimates from posterior distributions in [41]; we refer to this as

the deterministic model). This implicitly assumes that hosts are all the same or that parasites are moving between hosts sufficiently quickly that they adapt to an “average” environment.”

We also clarify our assumptions and aims with the semi-stochastic analysis, l. 271-275,

“Here we are asking, if parasites are “optimized” to the average host, does environmental and developmental variation change inferences about which cues are best? We quantified the effect of variation on cue performance using the geometric mean fitness, commonly used to assess fitness in fluctuating environments [59, 60].”

- I also would like to remark that the authors seem to not take into account the potential effect of temporal distribution of transmission events (being bitten by a vector) - it's assumed implicitly uniform over infection. This distribution could definitely have an effect on the optimal parasite strategy. So far, only the end of the infection is taken as the optimization horizon.

The reviewer is correct that we have assumed “the mosquito population is at a constant abundance and continuously biting, for the duration of the infection” (l. 642-643). In the Methods, we note that “[a]lthough this is a simplification of the evolutionary pressures faced by malaria parasites [35], it enables us to focus our investigation at the within-host level, where transmission investment “decisions” are being made and environmental cues are being perceived by parasites.” We note that we explored the effect of relaxing this assumption elsewhere (Greischar et al. 2019 *Evol*), but doing so necessitated a considerable simplification of another aspect of the biology, namely, in that work we assumed transmission investment was constant over the course of infections.

- It would be interesting, the authors mention it- but do not go deeper, to associate 'sensing' to a cost. The benefits of sensing and adapting their infection strategy, would have to be outweighing any costs for the parasite.

We agree with the reviewer that adaptive phenotypic plasticity is dependent on the benefit of plasticity outweighing the cost of perception. To that end, we have specified that one of the drawbacks of sensing multiple cues is the added metabolic burden (l. 404-406),

“Yet it is also widely appreciated that integrating multiple cues presents challenges, including the additional metabolic burden of sensing another cue [75]”

We have decided not to delve deeper into this topic, given the limited literature on the actual cost of cue sensing and the added complication that sensing different cues may have varying costs. Metabolic cost aside, we anticipate a bigger cost of sensing the “wrong” cue would be derived from a suboptimal transmission investment strategy, which we investigate in the manuscript.

- The authors mention their model choice is to depict immunologically naïve hosts. Why did they not simply vary initial conditions of their immunity variable to mimic the effect of immunological memory, in a similar way that they varied the other parameters? It seems straightforward, but maybe I am missing something. It would be good to clarify.

We decided to depict immunologically naive hosts due to the availability of immunological parameters for *Plasmodium chabaudi* in naive lab mice (Kamiya et al. 2020 *PLoS Comp Biol*). We have added additional clarifications about this choice in the Methodology (l. 522-525). We agree with the reviewer that understanding the impact of immunity on *Plasmodium* transmission investment evolution is important and could potentially be achieved by making assumptions about how immunological parameters might vary and rerunning the optimization algorithm. Given the consensus among reviewers that there was already too much in our manuscript, we think this is an interesting extension to pursue in the future, as mentioned in the discussion (l. 444-447).

- In the discussion it would be useful to clearly remark and emphasize which are the new insights on Plasmodium biology and which are broader novel insights obtained in this work.

We agree with the reviewer that the Discussion will benefit from streamlining and emphasizing further connections to the broader literature. To that end, we have made a number of changes. Here we highlight three. First, we provided additional information linking our modelling result on the benefit of perceiving parasite-centric cues to the broader, empirical literature (l. 377-381),

“studies have found that Plasmodium transmission investment varies in response to conspecific density [11, 12], parasite-conditioned media [14, 15], and infected red blood cell (iRBC)-derived vesicles [16], suggesting that parasites possess the ability to sense iRBC density. Our results support these responses being part of an adaptive strategy.”

Second, we rewrote the discussion section detailing the benefits and drawbacks of sensing multiple cues. To give readers a more intuitive understanding of why dual cue perception is beneficial, we provided a simplified explanation on why perceiving a single cue cannot allow for terminal investment (l. 387-390)

“Since most within-host variables go up and down, or vice versa, over time (e.g., iRBC densities rise and then fall due to resource limitation or immunity), one cue cannot provide enough detail to unambiguously establish “where” a parasite is in the course of infection. Combining cues solves this problem.”

Third, we expanded our discussion of sensing multiple cues in order to make connections to the broader literature on phenotypic plasticity (l. 399-410)

“There is a growing appreciation that, across diverse organisms, plasticity is often expressed in response to multiple cues that allow for more finely-tuned strategies [70–72]. Our work adds support to theories suggesting that the adaptive benefits of phenotypic plasticity are enhanced when cues provide non-redundant information [73] and enable organisms to sense distinct states that correspond to specific optimal phenotypes [74]. Yet it is also widely appreciated that integrating multiple cues presents challenges, including the additional metabolic burden of sensing another cue [75] and the multiplicative effect of noise when cues are non-additive [72]. Our complex (decidedly non-additive) dual cue reaction norms present a trade-off: they allow parasites to more accurately recreate optimal strategies in a given environment while rendering them more susceptible to stochastic environmental and developmental change, i.e., host-to-host variability.”